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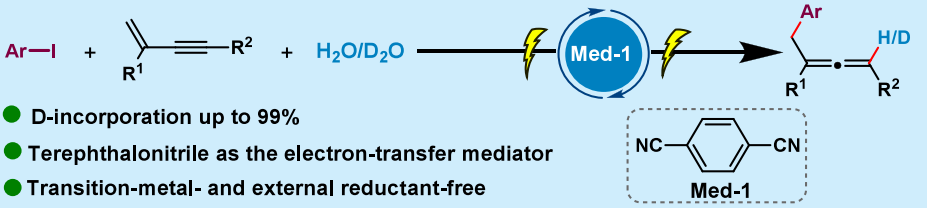
Electroreductive 1,4-Hydro(deutero)arylation of 1,3-Enynes Enabled by Terephthalonitrile: Entry to Trisubstituted Allenes

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ABSTRACT: An electroreduction method for regioselective 1,4-hydro(deutero)arylation of 1,3-enynes with aryl iodides and H_2O or D_2O has been developed, delivering diverse aryl-functionalized trisubstituted allenes under mild, transition-metal-free, and external reductant-free conditions. Upon the activation of terephthalonitrile as the organic electron transfer mediator, the protocol enables the formation of aryl radicals via single-electron transfer (SET) reduction with aryl iodides. The protocol shows a broad substrate scope with a good functional group compatibility and allows deuterium labeling using D_2O with up to 99% D-incorporation. Mechanistic studies support a cathodic radical anion−single electron reduction−cathodic radical polar crossover relay pathway, offering a sustainable approach to access trisubstituted allenes.

The regioselective functionalization of π-conjugated systems remains a central pursuit in modern synthetic chemistry, offering strategic routes to construct value-added architectures from simple and readily available precursors. Among these systems, 1,3-enynes, which are composed of conjugated alkene and alkyne moieties, serve as highly versatile synthons for the construction of complex molecules in synthesis.[1−3] In particular, the 1,4-hydrofunctionalization of 1,3-enynes has emerged as a powerful strategy for the direct construction of diverse functionalized allenes,[3−7] which are privileged structural motifs widely found in natural products, pharmaceuticals, and functional materials and are highly reactive synthetic intermediates.[8] Conventionally, 1,3-enyne 1,4-hydrofunctionalization approaches, such as hydrosilylation, hydroborylation, hydroamination, and hydrocarbonylation, have predominantly relied on the transition-metal catalysts, such as palladium, rhodium, copper, and nickel (Scheme 1A).[3,4] While these methods have achieved remarkable levels of reactivity and selectivity, they often require the use of highly expensive transition metals and stoichiometric external reductants and/or may generate toxic byproducts. Recently, radical-mediated 1,3-enyne 1,4-hydrofunctionalizations have been extensively developed, providing access to a wide range of functionalized allenes (Scheme 1B).[6,7] These reactions typically proceed through photoredox, electrochemical, or nickel reductive catalysis to generate radicals, followed by radical addition to the enyne, single-electron reduction, and subsequent protonation. For example, the 1,4-hydroalkylation of 1,3-enynes with alkyl bromides or alkyl *N*-hydroxyphthalimide (CONHPI) esters involves single-electron transfer (SET) reduction of the alkyl halide by Ni(0) or Ni(I), generating alkyl radicals that furnish trisubstituted allenes via a nickel-catalyzed reductive radical−polar crossover (RPC) pathway (Scheme 1B-b).[7] Despite that radical-mediated 1,4-hydrofunctionalizations such as hydroborylation, hydro(deutero)silylation, hydrofluoromethylation, and hydrophosphinylation have been well established, the 1,4-hydroarylation of 1,3-enynes to access highly valuable aryl-substituted allenes, to the best of our knowledge, remains unexplored.[6] Additionally, the pursuit of greener alternatives continues to drive the development of transition-metal-free radical chemistry, which opens new avenues for modular allene synthesis, while overcoming key limitations of traditional metal-catalyzed protocols in terms of sustainability, cost, and functional group compatibility.

Driven by this goal, we initiated our investigation into the hydroarylation of 1,3-enynes with aryl halides under electrochemical conditions (Scheme 1C). Our strategy centers on the electroreductive generation of aryl radicals via cathodic single-electron transfer (SET) from aryl halides, facilitated by organic redox mediators,[9] thereby avoiding the need for metal

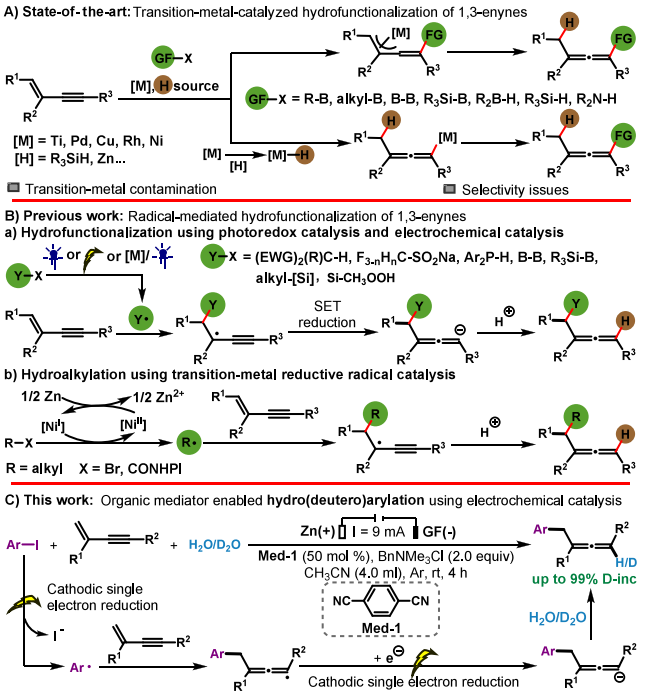
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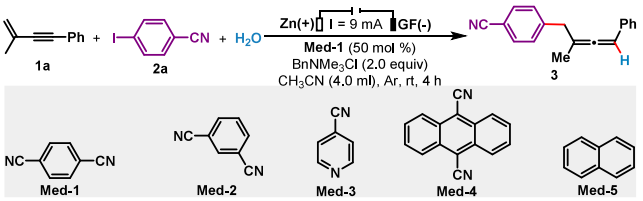
Scheme 1. Hydrofunctionalization of 1,3-Enynes



catalysts. We surmised that the *in situ* generated aryl radicals would undergo selective addition to the distal (alkene) carbon of 1,3-enyne, forming a short-lived allenyl radical intermediate. A subsequent SET reduction and protonation of this intermediate would then deliver the desired allenes with high regioselectivity. Beyond avoiding transition metals, this mediated electrolysis strategy offers several key advantages: it circumvents the high reduction potential of aryl halides, enabling aryl radical generation under mild electrochemical conditions, and prevents degradation of sensitive functional groups often observed under high voltage. Additionally, water can serve as a benign and abundant proton source, and the method is readily adaptable to use low-cost and commercially available D_2O as a deuterium donor, enabling the efficient synthesis of deuterium-labeled allenes—molecules of growing value in pharmaceutical, metabolic, and mechanistic studies.[10]

We commenced our study by evaluating optimal conditions for 1,4-hydroarylation using (3-methylbut-3-en-1-yn-1-yl)-benzene **1a** and 4-iodobenzonitrile **2a** as the model substrates (Table 1). After extensive optimization, we identified effective conditions that involve a combination of an organic electron transfer mediator terephalonitrile **Med-1**, H_2O (50 equiv) as the proton source, benzyltrimethylammonium chloride ($BnNMe_3Cl$) as the supporting electrolyte, and MeCN as the medium in an undivided cell equipped with a zinc cathode and a graphite felt (GF) anode at 9 mA constant current, affording the desired allene product **3** in a 65% isolated yield (entry 1). The results show that the constant current, mediator **Med-1**, and $BnNMe_3Cl$ electrolyte are essential for the reaction success, as no desired 1,4-hydroarylation reaction occurred when lacking each (entries 2, 7, and 13). Replacement of **Med-1** by other organic mediators **Meds-2–5** resulted in significantly lower yields or no formation of the desired product (entries 3–5), further underscoring the importance of mediator structures. Reducing the **Med-1** loading to 10 mol % led to a decreased yield (entry 6). Other electrolytes, such as $BnNEt_3Cl$, Et_4NCl, and $^nBu_4NCl$,

Table 1. Screening of the Optimal Reaction Conditions[a]



entry	variation from the standard conditions	yield (%)
1	none	65
2	without **Med-1**	0
3	using **Med-2**	21
4	using **Med-3**	28
5	using **Med-4** or **Med-5**	trace
6	with 10 mol % **Med-1**	38
7	without $BnNMe_2Cl$	0
8	$BnNEt_2Cl$ instead of $BnNMe_3Cl$	41
9	$^nBu_4NCl$ instead of $BnNMe_2Cl$	50
10	Et_4NBF_4 instead of $BnNMe_2Cl$	39
11	using Zn(+)/C rod (−)	53
12	using Zn(+)/Pt(−)	36
13	without current	0
14	6 mA instead of 9 mA	54
15	12 mA instead of 9 mA	65
16	DMF or DCM instead of DMF	trace
17[b]	none	60

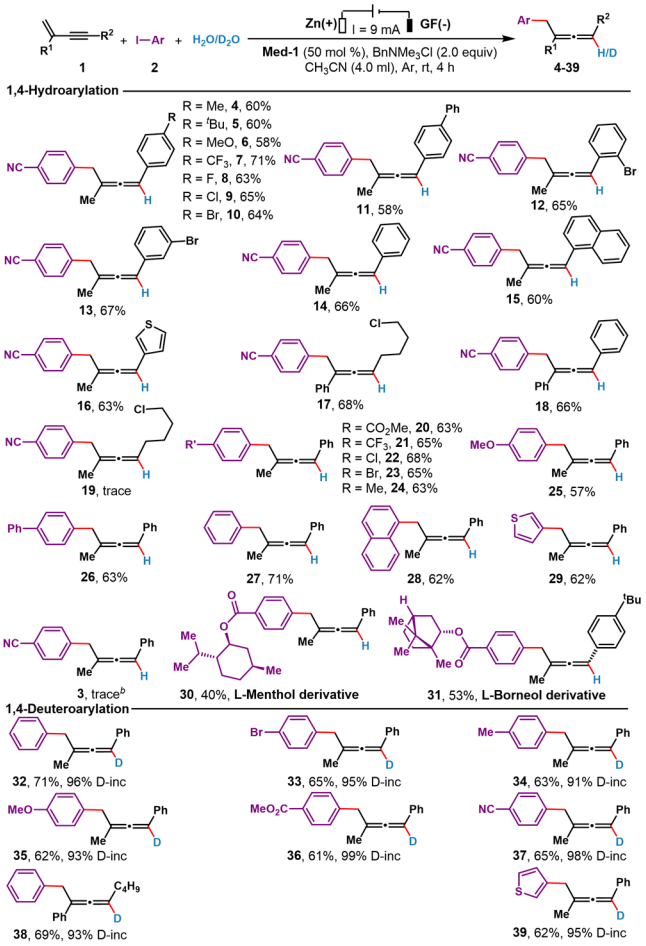
[a]Standard reaction conditions: undivided cell, zinc flake anode (15 × 30 × 0.1 mm), GF cathode (15 × 30 × 0.2 mm), constant current = 9 mA, **1a** (0.2 mmol), **2a** (0.3 mmol; 1.5 equiv), H_2O (50 equiv), **Med-1** (50 mol %), $BnNMe_3Cl$ (2 equiv), MeCN (4 mL), argon, and room temperature. GF = graphite felt. Med = mediator. [b]**1a** (1 mmol). See the Supporting Information for details.

led to inferior results (entries 8–10). The combination of a zinc cathode and graphite felt anode along with a constant current of 9 mA proved optimal, and other electrode materials or current settings reduced efficiency (entries 11–15). Among examinations of the solvent effect (such as MeCN, DMF, and DCM), the results revealed that MeCN was found to be the best solvent option (entries 1 and 16). Delightedly, the reaction could be handled with a scale up to 1 mmol **1a**, providing **3** in good yield (entry 17).

After the optimized reaction conditions were established, the substrate scope of this electrochemical 1,3-enyne 1,4-hydro-(deutero)arylation protocol was assessed by reaction of a diverse array of 1,3-enynes **1** with aryl halides **2** in the presence of H_2O under the optimized conditions (Table 2). Initially, various structurally and electronically varied 4-aryl-substituted 1,3-enynes, including electron-rich and electron-deficient aryl-substituted ones, were examined and showed high reactivity for this electrochemical 1,3-enyne 1,4-hydroarylation, affording allenes **4–18** in moderate to good yields. A wide range of functionalities, including Me, ^{t}Bu, MeO, CF_3, F, Cl, and Br, on the aryl ring at the terminal alkyne were well tolerated (**4–13**). Especially, halo groups such as F, Cl, and Br remained intact during the hydroarylation process, thereby offing highly reactive potential handles for further derivatizations. Notably, 4-aryl 1,3-enynes bearing a Br group at the *para*, *meta*, and *ortho* position, respectively, underwent the 1,4-hydroarylation smoothly, affording **10** and **12–13** in good yields. The results indicated that the steric hindrance did not affect the reaction, and the reaction might be applicable to aryl bromides. Indeed,

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Table 2. Scope of 1,3-Enynes (1) and Aryl Halides (2)[a]



[a]Reaction conditions: Undivided cell, zinc flake anode (15 × 30 × 0.1 mm), GF cathode (15 ×30 × 0.2 mm), constant current = 9 mA, **1** (0.2 mmol), **2** (0.3 mmol; 1.5 equiv), H_2O (50 equiv), **Med-1** (50 mol %), $BnNMe_3Cl$ (2 equiv), MeCN (4 mL), argon, and room temperature. [b]4-CNC_6H_4Br (0.2 mmol).

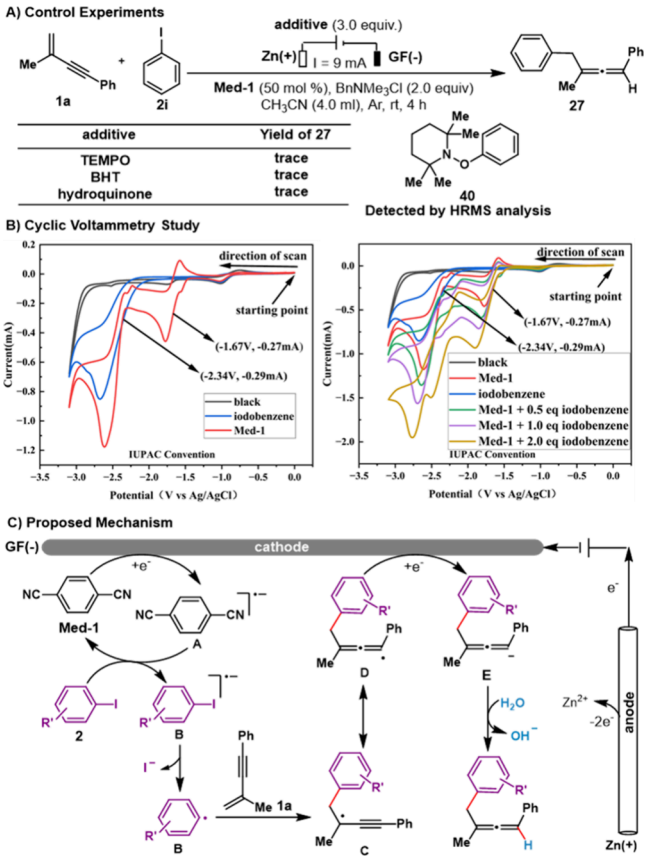
attempts to execute the 1,4-hydroarylation of 1,3-enyne **1a** and 4-bromobenzonitrile failed. Extended aromatic systems such as biphenyl, naphthalene, and thiophene, containing 1,3-enynes, were also compatible for furnishing allenes **11** and **15**−**16** in 58−63% yields. Interestingly, 4-alkyl 1,3-enyne was a suitable substrate, giving **17** in a good yield. The 2,4-diaryl-substituted 1,3-enyne was well tolerated, affording the product in good yield (**18**), whereas the 2,4-dialkyl analogue gave no product (**19**).

Next, the scope of aryl halides was investigated in the presence of (3-methylbut-3-en-1-yn-1-yl)benzene **1a**, H_2O, **Med-1**, and $BnNMe_3Cl$ (**20**−**31**). The reaction demonstrated excellent functional group compatibility: a wide range of electron-deficient and electron-rich aryl iodides bearing a functionality, such as CO_2Me, CF_3, Cl, Br, Me, MeO, and Ph, on the aryl ring efficiently manipulated 1,4-hydroarylation with 1,3-enyne **1a**, affording **20**−**26** in moderate to good yields. Electron-neutral iodobenzene also accommodated this 1,4-hydroarylation (**27**). Fortunately, aryl iodides possessing important naphthalene and thiophene functionalities were found to be compatible with the optimized conditions, giving allenes **28**−**29** in good yields. The regioselectivity and functional group tolerance of the protocol were further demonstrated by the successful 1,4-hydroarylation of 1,3-enynes with a methanol L-moiety (**30**) or an L-borneo-derived aryl iodide (**31**), underscoring the applicability of this method to construct complex and biologically relevant molecules.

Encouraged by these results, we further extended this electroreductive strategy to 1,4-deuteroarylation of 1,3-enynes using terephalonitrile **Med-1** as the organic electron transfer mediator (**32**−**39**).[10] An array of aryl iodides, including C_6H_5I, 4-BrC_6H_4I, 4-MeC_6H_4I, 4-$MeOC_6H_4I$, 4-$CO_2MeC_6H_4I$, 4-CNC_6H_4I, and 3-iodothiophene, were competent to operate the 1,4-deuteroarylation of 1,3-enyne **1a**, delivering the corresponding deuterium-labeled allenes **32**−**37** and **39** in good yields with high deuterium incorporation (91−99% D). Gratifyingly, aliphatic 1,3-enyne also accommodated this 1,4-deuteroarylation, affording the deuterium-labeled allene **38** in 69% yield with 93% deuterium incorporation.

To gain further insight into the reaction mechanism, a series of control mechanistic experiments were carried out (Scheme 2). The addition of radical scavengers such as TEMPO, BHT, and hydroquinone completely suppressed allene formation, suggesting the involvement of radical intermediates in the 1,4-hydroaryl process. Furthermore, the phenyl radical, generated from iodobenzene, was successfully trapped by TEMPO, and the corresponding adduct **40** was detected by high-resolution mass spectrometry (HRMS), providing direct evidence for aryl radical formation (Scheme 2A). Cyclic voltammetry (CV) studies revealed that the reduction potential of iodobenzene is significantly more negative than that of **Med-1**, indicating that **Med-1** is preferentially reduced at the cathode (Scheme 2B).[7]

Scheme 2. Control Experiments, Cyclic Voltammetry Study, and Possible Reaction Mechanism



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Furthermore, a continuous increase in the catalytic current of **Med-1** was observed upon the addition of iodobenzene, consistent with a single-electron transfer (SET) process from the reduction form of **Med-1** (dicyanobenzene radical anion) to the aryl iodide (Scheme 2B).

Based on these observations and the precedent literature,[1−7,9] a plausible mechanism was proposed as illustrated in Scheme 2C. Initially, terephthalonitrile **Med-1** undergoes SET reduction at the cathode to generate the terephthalonitrile radical anion intermediate **A**.[9] This activated species **A** then operates SET reduction with aryl iodide **2** to generate the aryl iodide radical anion intermediate **B**, followed by the C−I bond cleavage to afford the aryl radical **C**. Chemoselectivity addition of the aryl radical **C** across the C═C bond of 1,3-enyne **1** gives rise to the formation of the propargyl radical **C** and then the allene radical intermediate **D**.[6,7,11] Cathodic SET reduction of the intermediate **D** delivers the allene carbanion intermediate **D**.[5] Finally, protonation of intermediate **D** with H_2O furnishes the desired allene product and the OH^- ion, species, which upon protonation by water, furnish the final allene product.

In summary, we have developed a transition-metal-free electrochemical strategy for the regioselective electroreductive 1,4-hydro(deutero)arylation of 1,3-enynes with aryl iodides and H_2O/D_2O, enabling efficient access to trisubstituted allenes under mild conditions. The use of an organic redox mediator overcomes the high reduction potential of aryl halides, facilitating aryl radical formation via SET reduction. The protocol features a broad substrate scope and a high functional group tolerance, which is compatible with D_2O, allowing for the synthesis of deuterium-labeled allenes. Mechanistic studies support a cathodic radical anion single-electron-reduction cathodic radical polar crossover relay pathway, highlighting the potential utilities of this sustainable and versatile protocol.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c03768.

General experimental procedures, characterization details, and NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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